Research Methodology
Course Progress 44.72% (11 of 12)

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Assessment

Week 12 : Assignment

The due date for submitting this assignment has passed. Your last recorded submission was on 2026-10-12, 19:54 IST.

Due date on 2026-10-14, 23:59 IST

1. Which of the following statements correctly distinguish course-based education from research-based education as discussed in the lecture? 1 Point

- [x] Course-based education primarily deals with an existing body of knowledge, whereas PhD research is expected to create new knowledge.
- [ ] A PhD programme follows a common prescribed syllabus and curriculum in the same way as a course-based degree.
- [x] Course-based education generally has a prescribed syllabus and curriculum, whereas research-based education does not simply follow a fixed syllabus.
- [ ] Knowledge acquired in course-based education and new knowledge generated through PhD research are evaluated in exactly the same way.

Yes, the answer is correct. Score: 1

Accepted answer is: Course-based education primarily deals with an existing body of knowledge, whereas PhD research is expected to create new knowledge., Course-based education generally has a prescribed syllabus and curriculum, whereas research-based education does not simply follow a fixed syllabus.

2. A research scholar is beginning a PhD and expects the research guide to provide a complete step-by-step procedure for the entire research programme. Which responses are consistent with the lecture? 1 Point

- [x] The scholar should recognize that PhD research involves substantially self-directed work.
- [ ] The absence of a fixed syllabus means that the scholar does not need guidance from the research guide.
- [x] The research guide can provide guidance and support when ambiguity or difficulty arises.
- [x] The scholar is expected to develop the ability to decide how the research should progress.

Yes, the answer is correct. Score: 1

Accepted answer is: The scholar should recognize that PhD research involves substantially self-directed work., The research guide can provide guidance and support when ambiguity or difficulty arises., The scholar is expected to develop the ability to decide how the research should progress.

3. A researcher proposes a hypothesis based on existing knowledge and experience. Which conditions, according to the lecture, are relevant to treating it as a scientific hypothesis? 1 Point

- [x] It should be an educated prediction rather than a statement already established as true.
- [x] It should allow a possibility of being disproved.
- [ ] It should be protected from contradictory observations by adding explanations that cannot be tested.
- [ ] Its truth must be established before experiments or data collection begin.

Yes, the answer is correct. Score: 1

Accepted answer is: It should be an educated prediction rather than a statement already established as true., It should allow a possibility of being disproved.

4. Consider the hypothesis: "All swans are white." Based on the falsifiability example in the lecture, which statements are correct? 1 Point

- [x] Finding a black swan would provide an observation capable of disproving the hypothesis.
- [ ] It is necessary to observe every swan in the world before the hypothesis can be scientifically examined.
- [x] The possibility of disproving the hypothesis is central to the criterion discussed in the lecture.
- [x] The criterion discussed emphasizes the possibility of disproof rather than establishing the hypothesis as permanently true.

Yes, the answer is correct. Score: 1

Accepted answer is: Finding a black swan would provide an observation capable of disproving the hypothesis., The possibility of disproving the hypothesis is central to the criterion discussed in the lecture., The criterion discussed emphasizes the possibility of disproof rather than establishing the hypothesis as permanently true.

5. A student starts a PhD after being given a broad research topic. According to the lecture, which actions are appropriate before arriving at a testable hypothesis? 1 Point

- [x] Begin by formulating relevant questions about the topic and considering why the topic is important.
- [ ] Start an unrestricted literature survey immediately and use it as the only route for identifying the research problem.
- [x] Use literature exploration and observations to refine the initial questions through repeated cycles.
- [ ] Formulate the final testable hypothesis immediately from the broad topic without further exploration.

Yes, the answer is correct. Score: 1

Accepted answer is: Begin by formulating relevant questions about the topic and considering why the topic is important., Use literature exploration and observations to refine the initial questions through repeated cycles.

6. In the obesity and eating-habit example, observations from the literature and study were used to identify an anomaly and develop a hypothesis. Which statements are supported by the lecture? 1 Point

- [x] An anomaly can arise when observations do not match a commonly assumed explanation.
- [x] The observed influence of factors such as the amount of food offered can motivate further investigation.
- [ ] The hypothesis was formulated directly from the broad topic without literature survey or observations.
- [x] Identifying a missing link or contradiction can provide scope for further research and help generate a hypothesis.

Yes, the answer is correct. Score: 1

Accepted answer is: An anomaly can arise when observations do not match a commonly assumed explanation., The observed influence of factors such as the amount of food offered can motivate further investigation., Identifying a missing link or contradiction can provide scope for further research and help generate a hypothesis.

7. According to the Revised Bloom's Taxonomy discussed in the lecture, which statements are correct? 1 Point

- [x] Remember, Understand and Apply are described as lower-order thinking levels.
- [x] Analyze, Evaluate and Create are described as higher-order thinking levels.
- [ ] Create is placed below Remember because generating a new idea generally requires less cognitive effort.
- [ ] The taxonomy states that every individual experiences exactly the same difficulty across all six levels.

Yes, the answer is correct. Score: 1

Accepted answer is Remember, Understand and Apply are described as lower-order thinking levels., Analyze, Evaluate and Create are described as higher-order thinking levels.

8. A PhD scholar collects experimental data, examines unfamiliar patterns, considers alternative hypotheses, and develops a new research idea. Which Bloom's levels are most directly represented by these activities? 1 Point

- [x] Analyze
- [x] Evaluate
- [ ] Remember
- [x] Create

Yes, the answer is correct. Score: 1

Accepted answer is Analyze, Evaluate, Create

9. Which statements correctly reflect the lecture's discussion of the role of higher-order thinking in PhD research? 1 Point

- [x] All six levels of thinking may be used during research problem solving.
- [ ] The lecture identifies analysis as an important starting level in PhD research because researchers collect and analyze data.
- [ ] The main intellectual focus of a PhD is memorizing information from the existing body of knowledge.
- [x] Evaluation can involve considering alternate hypotheses and attempting to validate them.

Partially Correct. Score: 0.67

Accepted answer is: All six levels of thinking may be used during research problem solving., The lecture identifies analysis as an important starting level in PhD research because researchers collect and analyze data., Evaluation can involve considering alternate hypotheses and attempting to validate them.

10. Which statements are consistent with the lecture's discussion of creativity and functional fixedness? 1 Point

- [x] Creative thinking can be developed or improved through practice rather than being limited to an ability fixed at birth.
- [x] Functional fixedness can cause a person to perceive an object only through its conventional purpose and can hinder lateral thinking.
- [ ] Complete knowledge of the existing body of knowledge guarantees the creation of something new.
- [ ] Functional fixedness helps researchers generate new approaches when facing unexpected or novel challenges.

Yes, the answer is correct. Score: 1

Accepted answer is: Creative thinking can be developed or improved through practice rather than being limited to an ability fixed at birth., Functional fixedness can cause a person to perceive an object only through its conventional purpose and can hinder lateral thinking.